WORKED SOLUTION

# STEP II 2002, Question 13

Mechanics • Cdf

**STEP II 2002, Question 13**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q13

## The key idea

Although it sits in the mechanics section, this is a probability question about transforming a distribution function. Everything rests on one formula: differentiating $G$ gives $G'(y) = f(y) \cdot \dfrac{2}{(2-F(y))^2}$, where $f = F'$ is the density of $X$. So the density of $Y$ is the density of $X$ multiplied by a weight that is trapped between $\frac{1}{2}$ and 2. Once you have that, each expectation of $Y$ is an integral of $X$'s density times a weight in $[\frac{1}{2}, 2]$, and the inequalities follow by comparing integrands. The condition $a > 0$ is there to make the integrands non-negative.

## Solution

### Properties of $G$

We assume, as the question intends, that $X$ has a density $f(y) = F'(y) \geq 0$ on $[a, b]$, so that all of $X$'s probability lies in $[a, b]$.

Since $F(a) = 0$ and $F(b) = 1$,

$$G(a) = \frac{0}{2-0} = 0, \qquad G(b) = \frac{1}{2-1} = 1.$$

By the quotient rule,

$$G'(y) = \frac{F'(y)(2-F(y)) - F(y)(-F'(y))}{(2-F(y))^2} = \frac{2F'(y)}{(2-F(y))^2}.$$

The numerator is non-negative, because a distribution function is non-decreasing, so $F'(y) \geq 0$. The denominator is a square of $2 - F(y) \geq 1$, hence positive. So $G'(y) \geq 0$.

For the bound, $0 \leq F(y) \leq 1$ because $F$ is a distribution function. So

$$1 \leq 2 - F(y) \leq 2 \quad \Longrightarrow \quad 1 \leq (2-F(y))^2 \leq 4 \quad \Longrightarrow \quad \tfrac{1}{2} \leq \frac{2}{(2-F(y))^2} \leq 2,$$

where squaring keeps the inequalities because everything is positive, and taking reciprocals and multiplying by 2 reverses and then rescales them. This is the required double inequality.

Together, $G(a) = 0$, $G(b) = 1$ and $G$ non-decreasing mean that $G$ is a genuine distribution function on $[a, b]$, so the random variable $Y$ makes sense. Its density is

$$g(y) = G'(y) = w(y)\, f(y), \qquad \text{where } w(y) = \frac{2}{(2 - F(y))^2} \in [\tfrac{1}{2}, 2].$$

**Bounds on** $\mathrm{E}(Y)$

**Ask yourself** Why can we multiply the inequality $\frac{1}{2} \le w(y) \le 2$ by $yf(y)$ and keep its direction?

Because $yf(y) \ge 0$ on $[a, b]$: we have $f(y) \ge 0$, and $y \ge a > 0$. This is exactly where the hypothesis $a > 0$ is used. Hence

$$\tfrac{1}{2}\, yf(y) \le y\, w(y) f(y) \le 2\, yf(y) \qquad \text{for } a \le y \le b,$$

and integrating over $[a, b]$ preserves the inequalities:

$$\tfrac{1}{2} \int_a^b yf(y)\, \mathrm{d}y \;\le\; \int_a^b y\, g(y)\, \mathrm{d}y \;\le\; 2 \int_a^b yf(y)\, \mathrm{d}y,$$

that is,

$$\tfrac{1}{2}\, \mathrm{E}(X) \le \mathrm{E}(Y) \le 2\, \mathrm{E}(X),$$

as required.

**Bound on** $\mathrm{Var}(Y)$

In the same way, $y^2 f(y) \ge 0$ (this needs only $f \ge 0$), so multiplying $w(y) \le 2$ by it and integrating gives

$$\mathrm{E}(Y^2) = \int_a^b y^2\, w(y) f(y)\, \mathrm{d}y \le 2 \int_a^b y^2 f(y)\, \mathrm{d}y = 2\, \mathrm{E}(X^2).$$

For the subtracted term we need a *lower* bound on $(\mathrm{E}(Y))^2$. From the previous part, $\mathrm{E}(Y) \ge \frac{1}{2}\mathrm{E}(X)$, and $\mathrm{E}(X) > 0$ since $X \ge a > 0$. Both sides are positive, so squaring is safe:

$$(\mathrm{E}(Y))^2 \ge \tfrac{1}{4}(\mathrm{E}(X))^2.$$

Therefore

$$\begin{aligned} \mathrm{Var}(Y) &= \mathrm{E}(Y^2) - (\mathrm{E}(Y))^2 \\ &\le 2\, \mathrm{E}(X^2) - \tfrac{1}{4}(\mathrm{E}(X))^2 \\ &= 2(\mathrm{Var}(X) + (\mathrm{E}(X))^2) - \tfrac{1}{4}(\mathrm{E}(X))^2 \\ &= 2\, \mathrm{Var}(X) + \tfrac{7}{4}(\mathrm{E}(X))^2, \end{aligned}$$

using $\mathrm{E}(X^2) = \mathrm{Var}(X) + (\mathrm{E}(X))^2$ in the third line. This is the required bound.

## Checks

Take $X$ uniform on $[1,3]$, so $F(y) = \frac{1}{2}(y-1)$ and $\mathrm{E}(X) = 2$. Since $Y \geq 1$, $\mathrm{E}(Y) = 1 + \int_1^3 (1 - G(y))\,\mathrm{d}y$; substituting $u = F(y)$ gives

$$\mathrm{E}(Y) = 1 + 2\int_0^1 \frac{2(1-u)}{2-u}\,\mathrm{d}u = 1 + 4(1 - \ln 2) \approx 2.227,$$

comfortably between 1 and 4. Intuitively, $w$ is small where $F$ is small and large where $F$ is near 1, so $Y$ leans towards the top of the interval and $\mathrm{E}(Y) > \mathrm{E}(X)$ here. We also checked numerically, for 300 random power-law distributions on random intervals, that $\frac{1}{2}\mathrm{E}(X) \leq \mathrm{E}(Y) \leq 2\mathrm{E}(X)$ and the variance bound both hold; the observed ratios $\mathrm{E}(Y)/\mathrm{E}(X)$ lay between about 0.92 and 1.31.

## Common slips

- Forgetting why $a > 0$ matters: if $X$ could be negative, $yf(y)$ could be negative and multiplying by it would flip the inequality.
- Bounding $\mathrm{Var}(Y)$ using $\mathrm{E}(Y) \leq 2\mathrm{E}(X)$: the term $(\mathrm{E}(Y))^2$ is subtracted, so you need its lower bound, not its upper one.
- Trying to compare $\mathrm{Var}(Y)$ with $\mathrm{Var}(X)$ directly by integrating $(y-\mu)^2$ against $g$: the two variances use different means, so work with $\mathrm{E}(Y^2)$ instead.

## Beyond STEP

Writing $g = wf$ is a simple instance of a change of measure: the new distribution has a density (a *likelihood ratio*, or Radon–Nikodym derivative) with respect to the old one, and bounds on that ratio transfer directly to bounds on expectations. This idea underlies importance sampling in statistics and the change of probability measure used in financial mathematics. Grimmett and Welsh, *Probability: An Introduction*, covers distribution functions, densities and expectations at the right level to take this further.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.